

EXCEL SYLLABUS

1)Excel Basics (Foundation)

Topics:

Excel Interface (Ribbon, Tabs, Sheet, Workbook) , Rows, Columns, Cells , Data Entry & Editing , Cell Formatting (Number, Date, Currency)

Basic Formulas: SUM() , AVERAGE() , COUNT() , MIN() , MAX()

AutoFill & Flash Fill

2) Important Functions (Core Excel for Analyst)

Logical Functions -

IF() , IFS() , AND() , OR() , NOT()

Lookup Functions -

VLOOKUP() , HLOOKUP() , XLOOKUP() , INDEX() , MATCH()

Text Functions - LEFT() , RIGHT() , MID() , LEN() , TRIM() , CONCAT()

3) Data Cleaning & Preparation

Remove Duplicates

Text to Columns

Data Validation (Dropdown)

Conditional Formatting

Find & Replace

Sorting & Filtering

4) Pivot Tables

Pivot Table Create karna

Rows / Columns / Values samajhna

Grouping (Month, Year)

Pivot Charts , Slicers

5) Advanced Excel for Data Analytics

Power Query (ETL Process)

Power Pivot

Data Modeling

What-If Analysis

Goal Seek

Scenario Manager

KPI Dashboard Creation

6) Charts & Visualization

Bar Chart

Line Chart

Pie Chart

Combo Chart

Formatting Charts Professionally

Dashboard Design Principles

MICROSOFT CORPORATION – IS AN AMERICAN MULTINATIONAL TECHNOLOGY COMPANY THAT PRODUCES COMPUTER SOFTWARE , CONSUMER ELECTRONICS , PERSONAL COMPUTER , RELATED SERVICES .

#MICROSOFT OFFICE – IS A BUNCH OF DIFFERENT APPS OR TOOLS THAT ARE USED IN OFFICES AND CORPORATIONS.

IT IS FIRST ANNOUNCED BY BILL GATES ON 1 AUGUST , 1988 IN LAS VEGAS.

ALTHOUGH MS OFFICE COMES WITH MANY TOOLS , BUT THREE OF THEM ARE MOST POPULAR

- 1) MS EXCEL
- 2) POWERPOINT
- 3) WORD

SOME OTHER FAMOUS APPS OF MS OFFICE ARE OUTLOOK , ONEDRIVE , TEAMS , NOTE , SKYPE.

MICROSOFT EXCEL – IS A SPREADSHEET SOFTWARE CREATED BY MICROSOFT THAT IS USED TO ORGANIZE NUMBER AND DATA IN SHEET .

MS EXCEL PROVIDES MANY FORMULAS AND FEATURES THAT HELP TO DO CALCULATIONS IN SPREADSHEETS

USES OF MS EXCEL

- 1) DATA ENTRY
- 2) DATA MANAGEMENT
- 3) ACCOUNTING
- 4) FINANCIAL ANALYSIS
- 5) CHARTING, GRAPHING
- 6) PROGRAMMING
- 7) TIME MANAGEMENT
- 8) BUSINESS ANALYSIS
- 9) PEOPLE MANAGEMENT
- 10) MANAGING OPERATIONS
- 11) PERFORMING REPORTING
- 12) OFFICE ADMINISTRATION
- 13) STRATEGIC ANALYSIS
- 14) PROJECT MANAGEMENT

Ribbon → Top wala area jisme tools hote hain (Home, Insert, Data, etc.)

Tabs → Ribbon ke andar categories (Home, Insert, Formulas)

Worksheet (Sheet) → Jahan data likhte hain (grid format)

Workbook → Puri Excel file jisme multiple sheets hoti hain

Row → Horizontal line (1,2,3... numbers se identify hoti hai)

Column → Vertical line (A,B,C... letters se identify hoti hai)

Cell → Row + Column ka intersection (jaise A1, B3)

DATATYPES IN EXCEL –

Text, Number, Date , Time , Currency , Percentage, Logical (Boolean) , Error Values

1. SUM()

Numbers ka total nikalta hai.

Example:

A1 = 10

A2 = 20

A3 = 30

Formula:

=SUM(A1:A3)

Output → 60

2. AVERAGE()

Numbers ka average (mean) nikalta hai.

Example:

=AVERAGE(A1:A3)

Output → 20

3. COUNT()

Sirf numbers count karta hai.

Example:

A1 = 10

A2 = Text

A3 = 30

=COUNT(A1:A3)

Output → 2

4. MIN()

Sabse chota number batata hai.

Example:

=MIN(A1:A3)

Output → 10

5. MAX()

Sabse bada number batata hai.

Example:

=MAX(A1:A3)

Output → 30

AutoFill & Flash Fill - Pattern ko automatically fill karta hai , Pattern ko identify karke automatic fill karta hai (Ctrl + E)

LOGICAL CONDITIONS

1) IF () - The IF() function checks a condition and returns one value if the condition is TRUE and another value if the condition is FALSE.

SYNTAX =IF(logical_test, value_if_true, value_if_false)

If marks are in cell A1:

EXAMPLE =IF(A1>=40,"Pass","Fail")

If A1 is 50 → Output: Pass

If A1 is 30 → Output: Fail

2) IFS () - The IFS() function checks multiple conditions and returns the value for the first TRUE condition.

SYNTAX =IFS(condition1,value1, condition2,value2, ...)

If marks are in A1:

EXAMPLE =IFS(A1>=75,"Distinction",
A1>=60,"First Class",
A1>=40,"Pass",
A1<40,"Fail")

If A1 = 68 → Output: First Class , If A1 = 35 → Output: Fail

3) AND() Function - The AND() function returns TRUE only if all conditions are TRUE.

SYNTAX =AND(condition1, condition2, ...)

EXAMPLE =AND(A1>=40, B1>=40)

Returns TRUE only if both subjects are passed.

If one subject is less than 40 → FALSE

4) OR() Function - The OR() function returns TRUE if at least one condition is TRUE.

Syntax =OR(condition1, condition2, ...)

Example =OR(A1>=40, B1>=40)

Returns TRUE if at least one subject is passed.

Returns FALSE only if both are failed.

5) NOT() Function - The NOT() function reverses the logical value (TRUE becomes FALSE, FALSE becomes TRUE).

Syntax =NOT(logical)

Example =NOT(A1>=40)

If A1 = 50 → FALSE

If A1 = 30 → TRUE

Function	Use
IF()	Single condition decision
IFS()	Multiple conditions
AND()	All conditions must be TRUE
OR()	At least one condition TRUE
NOT()	Reverse logical value

Text Functions –

1. LEFT()

Definition:

Returns the specified number of characters from the left side of a text.

Syntax - LEFT(text, num_chars)

2. RIGHT()

Definition:

Returns the specified number of characters from the right side of a text.

Syntax – (RIGHT(text,num_chars)

3. MID()

Definition:

Extracts characters from the middle of a text string.

Syntax - MID(text, start_num, num_chars)

4. LEN()

Definition:

Counts the total number of characters in a text (including spaces).

Syntax- LEN(text)

5. UPPER()

Definition:

Converts text into uppercase letters.

Syntax – upper(text)

6. LOWER()

Definition:

Converts text into lowercase letters.

Syntax – lower(text)

7. PROPER()

Definition:

Capitalizes the first letter of each word.

Syntax – proper(text)

8. TRIM()

Definition:

Removes extra spaces from text except single spaces between words.

Syntax – trim(text)

9. CONCAT()

Definition:

Joins text from multiple cells.

Syntax – Concat(text1 , text2)

10. FIND()

Definition:

Finds the position of a character in text (case-sensitive).

Syntax – find(C , Text)

11. SEARCH()

Definition:

Finds position of text but not case-sensitive.

Syntax- search (v , Text)

12. REPLACE()

Definition:

Replaces part of text with another text.

Syntax =REPLACE("DATA",1,2,"MA")

LOOKUP FUNCTION

- 1) LOOKUP - LOOKUP is an Excel function used to search for a value in a single row or column and return the corresponding value from another row or column.

Syntax - LOOKUP(lookup_value, lookup_vector, result_vector)

- 2) VLOOKUP- (Vertical Lookup) is used to search for a value in the first column of a table and return a corresponding value from another column in the same row.

Syntax - VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

- 3) HLOOKUP() - (Horizontal Lookup) is used to search for a value in the first row of a table and return a value from a specified row in the same column.

Syntax - HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])

4. XLOOKUP() - is a modern Excel function used to search for a value in a range and return the corresponding value from another range.

It can replace VLOOKUP and HLOOKUP.

Syntax - XLOOKUP(lookup_value, lookup_array, return_array)

5. INDEX() - is an Excel function used to return the value of a cell in a specified position within a table or range.

Syntax - INDEX(array, row_num, column_num)

6. MATCH() - is used to find the position of a value in a range of cells.

Syntax - MATCH(lookup_value, lookup_array, match_type)

7. INDEX + MATCH - INDEX and MATCH together are used to perform advanced lookups by finding the position of a value using MATCH and then retrieving the corresponding value using INDEX.

Syntax - =INDEX(B2:B4, MATCH(102,A2:A4,0))

Remove Duplicates - Remove Duplicates is an Excel feature used to delete repeated (duplicate) values from a dataset and keep only unique records.

Steps to Remove Duplicates

1. Select the data range (table or column).
2. Go to the Data tab in Excel.
3. Click Remove Duplicates.
4. Choose the column(s) where you want to check duplicates.
5. Click OK.

Excel will remove duplicate rows and keep only one unique value.

Text To Columns - Text to Columns is an Excel feature used to split the text in one column into multiple columns based on a delimiter or fixed width.

Types of Text to Columns

1. Delimited

Text is separated using special characters like:

- Comma (,)
- Space
- Tab
- Semicolon (;)

2. Fixed Width

Text is split based on specific character positions.

Steps to Use Text to Columns

1. Select the column containing the text.
2. Go to the Data tab.
3. Click Text to Columns.
4. Choose Delimited or Fixed Width.
5. Select the delimiter (comma, space, etc.).
6. Click Finish.

Data Validation

Data Validation is an Excel feature used to control or restrict the type of data that users can enter in a cell.

It helps maintain accurate and consistent data.

Types of Data Validation

1. List (Dropdown)

Allows users to select values from a dropdown list.

2. Whole Number

Allows only whole numbers in a cell.

Example

Only numbers 1 to 100 are allowed.

3. Decimal

Allows only decimal numbers.

Example

Numbers between 0.5 and 10.5

4. Date

Restricts data to valid dates.

Example

Only dates between 01-01-2024 and 31-12-2024

5. Text Length

Limits the number of characters.

Example

Maximum 10 characters allowed.

Conditional Formatting - Conditional Formatting is an Excel feature used to automatically apply formatting (such as color, icons, or data bars) to cells based on specific conditions or rules.

It helps to identify patterns, trends, and important values in data easily.

Types of Conditional Formatting

1. Highlight Cells Rules

Highlights cells based on conditions.

Examples:

- Greater Than
- Less Than
- Between
- Equal To
- Duplicate Values

2. Top/Bottom Rules

Highlights top or bottom values in a dataset.

Examples:

- Top 10 items
- Bottom 10 items
- Above Average
- Below Average

3. Data Bars

Shows bar charts inside cells based on value size.

4. Color Scales

Changes cell color based on value range.

Sorting & Filtering –

1. Sorting in Excel

Definition

Sorting is the process of arranging data in a specific order such as alphabetical order, numerical order, or date order.

Types of Sorting -

1. Ascending Order

- A → Z (text)
- Smallest → Largest (numbers)
- Oldest → Newest (dates)

2. Descending Order

- Z → A (text)
- Largest → Smallest (numbers)
- Newest → Oldest (dates)

2. Filtering in Excel

Definition

Filtering is used to display only the data that meets specific conditions and hide the rest of the data.

Only matching data will be visible.

Pivot tables - Pivot Table is an Excel tool used to summarize, analyze, and organize large amounts of data quickly by grouping and calculating values such as sum, count, average .

It helps in creating reports and analyzing data easily.

Uses of Pivot Table

- Summarize large data
- Create reports
- Calculate totals and averages
- Analyze business data quickly

1. Rows

Rows in a Pivot Table are used to display data labels vertically and organize data into row categories.

Example:

If the Name field is placed in the Rows area, each name will appear as a separate row in the Pivot Table.

2. Columns

Columns in a Pivot Table are used to display data labels horizontally and categorize data across columns.

Example:

If the Year field is placed in the Columns area, each year will appear as a separate column.

3. Values

Values in a Pivot Table represent the numerical data that is summarized using calculations such as Sum, Count, Average, Maximum, or Minimum.

Example:

If the Sales field is placed in the Values area, Excel can calculate Sum of Sales.

4. Grouping (Month / Year)

Grouping in a Pivot Table is used to organize data into categories such as months, quarters, or years for easier analysis.

Example:

Date data like 10-01-2024, 15-02-2024 can be grouped into January, February, etc.

5. Pivot Charts

Pivot Chart is a graphical representation of Pivot Table data used to visualize summarized information in the form of charts such as bar charts, line charts, or pie charts.

6. Slicers

Slicers are interactive visual filters used in Pivot Tables to filter data quickly by clicking buttons for specific categories.

Example:

A slicer for City can allow users to filter data by Pune, Mumbai, or Nashik.

Power Query - is an Excel tool used to extract, transform, and load (ETL) data from different sources and prepare it for analysis.

It helps in cleaning, combining, and transforming data automatically.

1. Power Pivot

Definition

Power Pivot is an Excel feature used to analyze large datasets, create relationships between multiple tables, and perform advanced calculations using DAX (Data Analysis Expressions).

Example

You can connect Sales table and Customer table and analyze them together.

2. Data Modeling

Definition

Data Modeling is the process of organizing and connecting multiple data tables using relationships so that they can be analyzed together efficiently.

Example

Table 1 → Customers

Table 2 → Orders

Using Customer ID both tables can be connected.

3. What-If Analysis

Definition

What-If Analysis is an Excel feature used to change input values to see how those changes affect the results of formulas.

Example

If Sales price changes, Excel shows how profit will change.

4. Goal Seek

Definition

Goal Seek is a What-If Analysis tool used to find the input value required to achieve a specific result in a formula.

Example

If profit formula is:

$$\text{Profit} = \text{Sales} - \text{Cost}$$

Goal Seek can calculate how much sales are required to get ₹10,000 profit.

Steps

1. Data tab
2. What-If Analysis
3. Goal Seek

5. Scenario Manager

Definition

Scenario Manager is a What-If Analysis tool used to create and compare multiple sets of input values to see different possible outcomes.

Example

Scenario 1 → High Sales

Scenario 2 → Medium Sales

Scenario 3 → Low Sales

Excel compares results for all scenarios.

6. KPI Dashboard Creation

Definition

KPI Dashboard is a visual report in Excel used to track and display key performance indicators (KPIs) using charts, tables, and metrics.

Example KPIs

- Total Sales
- Profit
- Customer Growth
- Monthly Revenue

Dashboard tools used:

- Pivot Tables
- Charts
- Slicers
- Conditional Formatting

1. Bar Chart - A Bar Chart is a data visualization method that uses rectangular bars to compare values across different categories. The length of each bar represents the magnitude of the data.

Usage

- Comparing categories
- Sales comparison
- Performance comparison

2.Line Chart - A Line Chart is used to display trends and changes in data over time by connecting data points with a continuous line.

Usage

- Trend analysis
- Time-series data
- Growth tracking

3.Pie Chart - A Pie Chart is a circular chart divided into slices to show the proportion or percentage contribution of each category to the whole.

Usage

- Percentage distribution
- Market share analysis
- Budget allocation

4. Combo Chart - A Combo Chart combines two different chart types (such as bar and line) in a single visualization to compare multiple data series with different scales.

Usage

- Comparing different metrics
- Sales vs Profit
- Revenue vs Target

5. Formatting Charts Professionally - Chart formatting means customizing the chart appearance to make it clear, readable, and visually appealing for professional reporting.

Important Formatting Steps

1. Add Chart Title
2. Label X-axis and Y-axis
3. Use proper colors
4. Remove unnecessary gridlines
5. Add data labels
6. Use consistent fonts
7. Highlight important values

Example

Instead of a plain chart, add:

- Title → *Monthly Sales Performance*
- Axis labels → *Month / Revenue*
- Data labels → exact values on bars

6. Dashboard Design Principles - A Dashboard is a visual display of key metrics and data insights on a single screen to support quick decision-making.

Important Principles

1. **Keep it simple**
Avoid too many charts.
2. **Use meaningful charts**
Line → trends
Bar → comparison
3. **Use consistent colors**
4. **Highlight KPIs**

Example KPIs:

- **Total Sales**
 - **Revenue**
 - **Profit**
 - **Growth %**
5. **Interactive elements**
 - **Filters**
 - **Slicers**
 - **Dropdowns**
 6. **Logical layout**

Top → KPIs

Middle → Charts

Bottom → Detailed data

Absolute Reference - Absolute Reference is a type of cell reference in Excel where the row and column remain fixed (locked) even when the formula is copied to another cell. It is represented using the \$ (dollar) symbol.

Syntax - \$Column\$Row

Types of Cell References

Type	Example	Behavior
Relative Reference	A1	Changes when copied
Absolute Reference	\$A\$1	Never changes
Mixed Reference	\$A1 / A\$1	Only row or column fixed

It is used when a constant value must remain fixed in formulas.

Common cases:

- Tax rate
- Commission rate
- Percentage
- Fixed price
- Exchange rate

Relative Reference (Excel) - A Relative Reference is a cell reference in Excel that automatically changes when a formula is copied or dragged to another cell.

It does not use the \$ (dollar) symbol.

It is used when you want excel to automatically adjust the formula for each row or column .

Type

Example Behavior

Common uses:

- Multiplication
- Addition
- Calculations across rows
- Copying formulas

Relative vs Absolute Reference

Type	Example	Behavior
Relative	A1	Changes automatically
Absolute	\$A\$1	Never changes
Mixed	A\$1 / \$A1	Row or column fixed

COUNT Functions

1. COUNT() -

The COUNT() function counts the number of cells that contain numeric values.

Syntax - COUNT(range)

2. COUNTA() -

The COUNTA() function counts the number of non-empty cells (numbers, text, or symbols).

Syntax - COUNTA(range)

3. COUNTIF() -

The COUNTIF() function counts cells that meet a specific condition.

Syntax - COUNTIF(range,criteria)

4. COUNTIFS() -

The COUNTIFS() function counts cells that satisfy multiple conditions.

Syntax - COUNTIFS(range1,criteria1,range2,criteria2)

Function	Purpose
COUNT	-Counts numeric values
COUNTA	-Counts non-empty cells
COUNTIF	-Counts cells with one condition
COUNTIFS	-Counts cells with multiple conditions

SUM Functions (Excel)

SUM functions are used to add numbers in Excel.

They help calculate totals, category totals, and conditional totals, which are very important in data analysis and reporting.

1 SUM() -

The SUM() function adds a range of numbers and returns the total.

Syntax - SUM(number1, number2, ...)

2 . SUMIF() -

The SUMIF() function adds numbers in a range that meet a single condition.

Syntax- SUMIF(range, criteria, sum_range)

3.SUMIFS()-

The SUMIFS() function adds numbers based on multiple conditions.

Syntax - SUMIFS(sum_range, criteria_range1, criteria1, criteria_range2, criteria2)

SUMIF adds values based on one condition.

SUMIFS adds values based on multiple conditions.

IFERROR Function (Excel)

The IFERROR() function in Excel is used to handle errors in formulas.

If a formula produces an error (such as #DIV/0!, #N/A, #VALUE, #REF), the IFERROR function returns a custom value instead of the error.

Syntax - IFERROR(value, value_if_error)

Error	Meaning
#DIV/0!	Division by zero
#N/A	Value not found
#VALUE	Wrong data type
#REF	Invalid reference
#NAME	Unknown function

DATE FUNCTION –

The DATE function in Excel is used to create a valid date by combining year, month, and day values into a single date.

Syntax - DATE(year, month, day)

Arguments

1. year – The year of the date (e.g., 2026)
2. month – The month number (1 = January, 12 = December)
3. day – The day of the month (1–31)

Related Date Functions

Some commonly used Excel date functions:

- **TODAY()** – Returns current date
- **NOW()** – Returns current date & time
- **DAY()** – Extracts day from a date
- **MONTH()** – Extracts month
- **YEAR()** – Extracts year
- **EDATE()** – Adds months to a date
- **EOMONTH()** – Last day of month

1. DATE()

Purpose: Create a date using year, month, and day.

Syntax **=DATE(year, month, day)**

Example

Formula	Result
=DATE(2026,3,6)	06-Mar-2026

2. TODAY()

Purpose: Returns current date automatically.

Syntax **=TODAY()**

Example

Formula	Result
=TODAY()	Current date

Use when you want today's date automatically update.

3. NOW()

Purpose: Returns current date + current time.

Syntax =NOW()

Example

Formula	Result
=NOW()	06-Mar-2026 15:30

4. DAY()

Purpose: Extract day from a date.

Syntax =DAY(date)

Example

Date	Formula	Result
06-Mar-2026	=DAY(A1)	6

5. MONTH()

Purpose: Extract month number from a date.

Syntax =MONTH(date)

Example

Date	Formula	Result
06-Mar-2026	=MONTH(A1)	3

6. YEAR()

Purpose: Extract year from a date.

Syntax =YEAR(date)

Date	Formula	Result
06-Mar-2026	=YEAR(A1)	2026

7. EDATE()

Purpose: Add or subtract months from a date.

Syntax =EDATE(start_date, months)

Example

Formula	Result
=EDATE("06-Mar-2026",3)	06-Jun-2026

8. EOMONTH()

Purpose: Find last day of the month.

Syntax =EOMONTH(start_date, months)

Formula	Result
=EOMONTH("06-Mar-2026",0)	31-Mar-2026

9. DATEDIF()

Purpose: Find difference between two dates.

Syntax =DATEDIF(start_date,end_date,"Y")

Example

Formula	Result
=DATEDIF("01-Jan-2020","06-Mar-2026","Y")	6 years

10. WORKDAY()

Purpose: Find working day excluding weekends.

Syntax =WORKDAY(start_date,days)

Example

Formula	Result
=WORKDAY("06-Mar-2026",5)	Next working day

